



NS3-Tutorial

OFDMA Codebase



OFDMA

OFDMA -> MU transmission -> transmission over RUs.

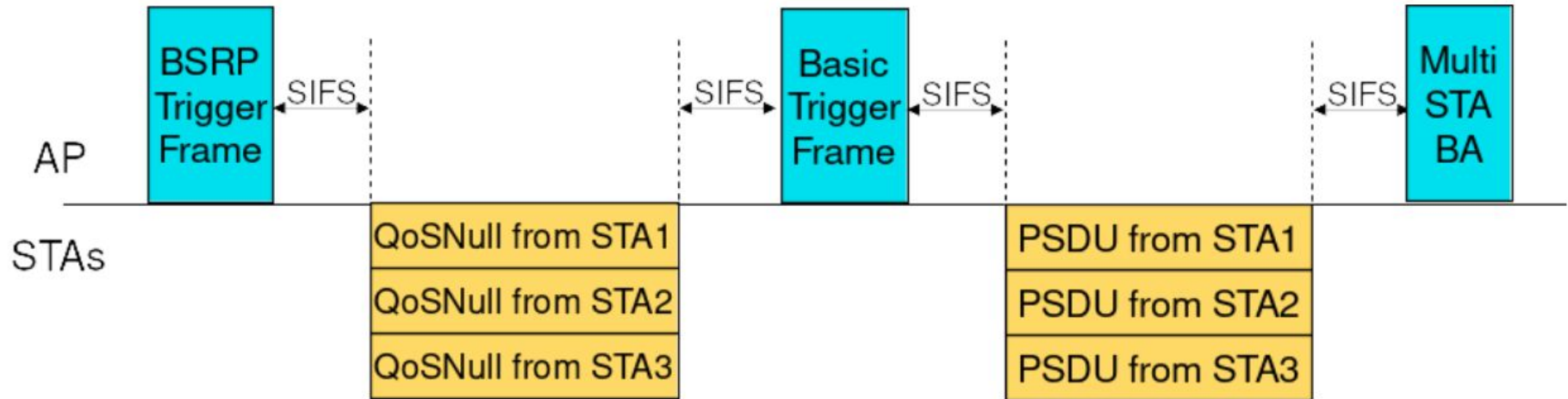
OFDM -> SU transmission -> transmission over complete channel.

DL -OFDMA



Acknowledgment of DL MU frames via aggregated MU-BAR Trigger Frames

UL-OFDMA



Frame exchange sequences using UL OFDMA



OFDMA -NS3

Example file in:

`examples/wireless/wifi-he-network.cc`



To run

Copy it to scratch folder.

```
./ns3 run "wifi-he-network.cc"    (will run with default parameters)
```

```
./ns3 run "wifi-he-network.cc --mcs=11"    (to provide parameters as command line arguments)
```

Can also be changed directly in the code.



Delay, path loss & fading

After the spectrum channel is defined:

```
Ptr<LogDistancePropagationLossModel> lossModel = CreateObject<LogDistancePropagationLossModel>();  
spectrumChannel->AddPropagationLossModel(lossModel);
```

```
Ptr<ConstantSpeedPropagationDelayModel> delayModel=CreateObject<ConstantSpeedPropagationDelayModel> ();  
spectrumChannel->SetPropagationDelayModel (delayModel);
```

```
Ptr<NakagamiPropagationLossModel> fadingModel = CreateObject<NakagamiPropagationLossModel> ();  
spectrumChannel->AddPropagationLossModel (fadingModel);
```



Minstrel Rate Adaptation

After “WifiHelper wifi” is defined

```
wifi.SetRemoteStationManager ("ns3::MinstrelHtWifiManager" );
```




Running for SU/OFDM & MU/OFDMA

Set the dlAckSeqType to “No-OFDMA” for SU

```
std::string dlAckSeqType{ "NO-OFDMA" };
```

Set the dlAckSeqType to “No-OFDMA” for SU

```
std::string dlAckSeqType{ "AGGR-MU-BAR" };
```

Can also be changed through command line

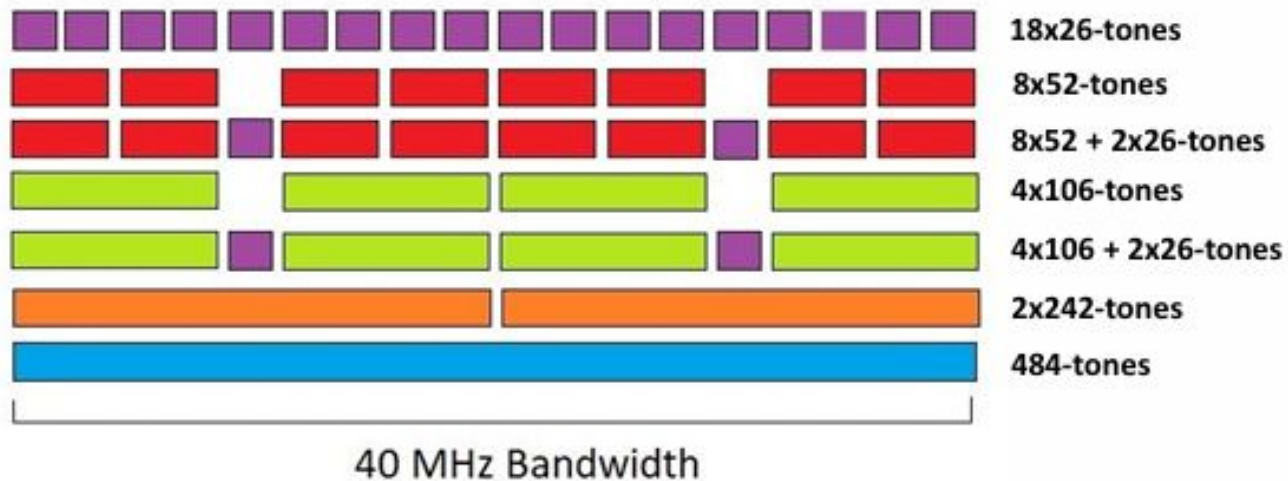
```
./ns3 run "wifi-he-network.cc --mcs=11 --dlAckType=AGGR-MU-BAR"
```



Running for UL OFDMA

Set `enableUlofdma` and `enableBsrp` to true.

RU Configurations





Varying RU configurations

RU configuration can be added to

```
mac.SetMultiUserScheduler ("ns3::RrMultiUserScheduler",  
                             "NStations", IntegerValue (m_maxNRus),  
                             "UseCentral26TonesRus", BooleanValue (m_useCentral26TonesRus));
```

Or in `rr-multi-user-scheduler.cc` directly.



Thank You